



Francisco Miguel Castro-Macías

Curriculum Vitae

Personal information

Full name Francisco Miguel Castro Macías
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Nationality Spain
Birth date 13/06/1998
Links [Personal webpage](#), [Google Scholar](#), [GitHub](#)

Professional summary

Research intern at Sony AI Zurich. PhD Candidate in Computer Science. Strong mathematical foundation (Double Degree in Mathematics & Computer Engineering). Experience in probabilistic Machine Learning, generative modeling, inverse problems, medical imaging, and molecular dynamics. Was a visiting researcher at the University of Cambridge (working on SDE-based MCMC samplers) and Northwestern University (working on attention-based mechanisms). Author of the [torchmil](#) PyTorch library. Published at NeurIPS, ICML, and ICIIP (Best Student Paper Award).

Education

2023–2026 **PhD. in Computer Science**, *Probabilistic machine learning methods for weakly supervised and inverse problems. Applications in medicine*, University of Granada, Granada.
2021–2022 **Msc. in Data Science and Computer Engineering**, University of Granada, Granada, GPA: 9.68/10.0.
2017–2021 **Degree in Mathematics**, University of Granada, Granada, GPA: 9.008/10.0.
2017–2021 **Degree in Computer Engineering**, University of Granada, Granada, GPA: 9.077/10.0.
Specialty in Computing and Intelligent Systems

Experience

- June 2026 – **Research intern**, Sony AI, Zurich, Switzerland.
Dec. 2026 Joined the Music Foundation Model team, working on generative modeling approaches for sound effects creation.
- Sep. 2025 – **Visiting PhD student**, Department of Engineering, University of Cambridge, United Kingdom.
Dec. 2025 Worked under the supervision of Prof. José Miguel Hernández-Lobato as a visiting student for three months. Extended the SDE-based formulation of diffusion models (including diffusion, flow matching, and stochastic interpolants) to develop a novel MCMC sampling method. Conducted experiments on particle and molecular systems. This work resulted in a paper accepted at ICML 2026.
- March 2024 – **Visiting PhD student**, Department of Electrical and Computer Engineering, Northwestern University, Evanston, Illinois, United States.
June 2024 Worked under the supervision of Prof. Aggelos K. Katsaggelos as a visiting student for three months. Conducted research on enhancing the localization capabilities of attention-based models (Transformers) by incorporating spatial information. Performed experiments on medical imaging datasets, including CT scans and whole slide images. This work resulted in a paper accepted at NeurIPS 2024.
- Jan. 2023 – **FPU PhD Student**, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.
Dec. 2026 Funded by an Formación Profesorado Universitario (FPU) research grant, awarded by Ministry of Universities of the Government of Spain.
- Jul. 2022 – **Research Assistant**, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.
Dec. 2022 Development and evaluation of methods for deblurring and color normalization of histological images

Training courses

- July 2025 **London Geometry and Machine Learning (LOGML) Summer School**, Imperial College, London.
Member of the project "Representation Learning and Knowledge Encoding with Biomedical Knowledge Graphs" led by Ruth Johnson. Contributed to expanding the vocabulary of a large language model (LLM) with biomedical knowledge tokens and conducted experiments during the fine-tuning phase.
- August 2022 **OxML Summer School: Health Track**, Mathematical Institute, University of Oxford, Oxford.
- April 2022 **Second training school Cloud Artificial Intelligence for Pathology (Clarify) project (EU funded)**, School of Computer Science and Telecommunications Engineering, University of Granada, Granada.

Grants

- Jan. 2023 – **Formación Profesorado Universitario (FPU)**, Department of Computer Science and Artificial Intelligence, University of Granada, Granada, Awarded by the Ministry of Universities of the Government of Spain.
Jan. 2027 This grant funded my PhD studies.

- Aug. 2022 **Blas Cabrera Summer School**, Universidad Internacional Menéndez Pelayo, Awarded by the Ministry of Universities.
One-week stay at the Palacio de la Magdalena, Santander. Attendance at seminars and talks given by the country's most renowned researchers
- Sep. 2020 – **Research Initiation Grant for Undergraduate Students**, *A federated learning model to predict COVID19 disease from chest x-rays images*, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.
July 2021 Extension of several deep learning models to the federated scenario. Theoretical and experimental study of various aggregators and data partitions. Under the supervision of Francisco Herrera Triguero and Siham Tabik Ouled Hrou.
- July 2017 **English language immersion (Level C1)**, Universidad Internacional Menéndez Pelayo, Huesca.
- July 2015 **Campus Científicos de Verano (Summer Science Campus)**, *TIC-TAC Robot*, Escuela Técnica Superior de Ingeniería, Departament d'Enginyeria Elèctrica, Electrònica i Automàtica, Universitat Rovira i Virgili, Tarragona.
- Sep. 2014 – **Secondary School Innovation Research Initiation Project in Andalusia**, *Stellar Clusters*, Consejo Superior de Investigaciones Científicas, University of Granada, Granada.
May 2015

Awards

- Nov. 2024 **Best Student Paper Award at 2024 IEEE International Conference on Image Processing (ICIP)**, *Bayesian Blind Image Deconvolution using a Hyperbolic-Secant prior*, [Link to the news](#).
- May 2022 **Best Final Degree Project in Computer Engineering**, School of Computer Science and Telecommunications Engineering, University of Granada, Granada.

Research

Publications

- 2026 **F. M. Castro-Macías**, P. Morales-Álvarez, S. Syed, D. Hernández-Lobato, R. Molina, J. M. Hernández-Lobato, *Conditional Diffusion Sampling*, 2026 International Conference on Machine Learning (ICML).
- 2026 **F. M. Castro-Macías**, F. J. Sáez-Maldonado, P. Morales-Álvarez, R. Molina, *torchmil: A PyTorch-based library for deep Multiple Instance Learning*, Neurocomputing.
- 2026 **F. M. Castro-Macías**, P. Morales-Álvarez, Y. Wu, R. Molina, A. K. Katsaggelos, *Probabilistic smooth attention for deep Multiple Instance Learning in medical imaging*, Pattern Recognition.
- 2025 **F. M. Castro-Macías**, M. Vega, R. Molina, *Deep Sparse Prior: Enhancing Deblurring Deep Priors with Sparsity Constraints*, 2025 IEEE International Conference on Image Processing Workshops (ICIP).
- 2025 **F. M. Castro-Macías**, F. Pérez-Bueno, M. Vega, R. Molina, A. K. Katsaggelos, *Deep Bayesian Blind Color Deconvolution using a Gaussian Scale Mixture Prior*, 2025 European Signal Processing Conference (EUSIPCO).

- 2024 **F. M. Castro-Macías**, F. Pérez-Bueno, M. Vega, J. Mateos, R. Molina, A. K. Katsaggelos, *Bayesian Blind Image Deconvolution using a Hyperbolic-Secant prior*, 2024 IEEE International Conference on Image Processing (ICIP), **Best Student Paper Award**.
- 2024 **F. M. Castro-Macías**, P. Morales-Álvarez, Y. Wu, R. Molina, A. K. Katsaggelos, *Sm-enhanced localization in Multiple Instance Learning for medical imaging classification*, 2024 Conference on Neural Information Processing Systems (NeurIPS).
- 2024 **F. M. Castro-Macías**, P. Morales-Álvarez, Y. Wu, R. Molina, A. K. Katsaggelos, *Hyperbolic Secant representation of the logistic function: application to Probabilistic Multiple Instance Learning for CT intracranial hemorrhage detection*, Artificial Intelligence Journal.
- 2024 **F. M. Castro-Macías**, F. Pérez-Bueno, M. Vega, J. Mateos, R. Molina, A. K. Katsaggelos, *Blind Color Deconvolution and Classification of Histological Images Using the Hyperbolic Secant Prior*, 2024 International Symposium on Biomedical Imaging (ISBI).
- 2023 S. Yang, F. Pérez-Bueno, **F. M. Castro-Macías**, R. Molina, A. K. Katsaggelos, *BCD-Net: Stain Separation of Histological Images using Deep Variational Bayesian Blind Color Deconvolution*, Digital Signal Processing.
- 2023 S. Yang, F. Pérez-Bueno, **F. M. Castro-Macías**, R. Molina, A. K. Katsaggelos, *Deep Bayesian Blind Color Deconvolution of histological images*, 2023 IEEE International Conference on Image Processing (ICIP).
- 2023 Y. Wu, **F. M. Castro-Macías**, P. Morales-Álvarez, R. Molina, A. K. Katsaggelos, *Smooth Attention for Deep Multiple Instance Learning: Application to CT Intracranial Hemorrhage Detection*, 2023 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI).

Reviewer experience

Served as a reviewer for top-tier machine learning venues: NeurIPS, ICML, TMLR, Transactions on Medical Imaging.

Teaching

Courses

- 2025 – 2026 **Algorithms**, Bachelor's Degree in Computer Engineering and Bachelor's Degree in Business Administration, University of Granada, Granada.
- 2024 – 2025 **Development and Design of Information Systems**, Bachelor's Degree in Computer Engineering, University of Granada, Granada.
- 2023 – 2024 **Development and Design of Information Systems**, Bachelor's Degree in Computer Engineering, University of Granada, Granada.
- 2023 – 2024 **ICT in Healthcare and Research Methodology**, Bachelor's Degree in Nursing, University of Granada, Granada.
- 2022 – 2023 **Image feature extraction (assistant)**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.

Bachelor's and Master's Thesis

- 2023 – 2024 **Blind Image Deconvolution using Deep Learning**, Bachelor's degree in Computer Engineering, University of Granada, Granada.
Student: J. A. Torres Coca, Supervisors: R. Molina, F. M. Castro-Macías.
- 2023 – 2024 **Uncertainty Estimation in Blind Image Deconvolution using Bayesian Deep Image Prior**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.
Student: A. Pérez Lara, Supervisors: R. Molina, F. M. Castro-Macías.
- 2022 – 2023 **Deep Multiple Instance Learning. Application to Intracranial Hemorrhage Detection**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.
Student: E. Jiménez Mariscal, Supervisors: P. Morales-Álvarez, F. M. Castro-Macías.

Talks

- 2026 **Introduction to Markov Chain Monte Carlo**, Bachelor's degree in Mathematics, Stochastic Processes Course.
- 2026 **Multiple Instance Learning for Histopathology: An Introduction to torchmil**, RISE MICCAI Tutorial Series, [Link to the video](#).
- 2025 **Denoising energy matching**, Visual Information Processing seminar.
- 2024 **The challenges of artificial intelligence in medicine**, Siete Colinas High School, Granada.
- 2024 **The challenges of artificial intelligence in medicine**, European Researchers' Night, Granada.
- 2024 **Medical Imaging Classification**, PyData, Granada.
- 2023 **The wisdom of crowds: Are many people smarter than one expert?**, European Researchers' Night, Granada.
- 2023 **The Attention Mechanism**, Visual Information Processing seminar.
- 2023 **Linearized Laplace Approximation for Deep Learning**, Visual Information Processing seminar.
- 2022 **Artificial Intelligence from the inside**, Academia El Futuro SLU, Granada.
- 2022 **That so-called Artificial Intelligence**, Centro de Estudios Hermanos Naranjo, Granada.

News and media

- Feb. 2025 **"Precision aid for locating cancer and tumours"**, IDEAL (Granada local newspaper). [Link](#).
- Dec. 2024 **"Visiting PhD Student Wins Best Student Paper Award at 2024 IEEE ICIP Conference"**, Newsletter of the Department of Electrical and Computer Engineering, Northwestern University. [Link](#).
- Nov. 2024 **"Francisco M. Castro Macías, awarded at the IEEE International Conference on Image Processing"**, Newsletter of the University of Granada. [Link](#).

———— Languages

Spanish Mother tongue
English Fluent

———— Sport achievements (Weightlifting)

Bronze Trophy "Feria de Almería de Halterofilia", -61Kg, 2019, Almería
Gold Spanish University Weightlifting Championships, -61Kg, 2019, Murcia
Silver Senior Weightlifting Championship of Andalucía, -61Kg, 2019, Cádiz